Cheatsheet AZ-305 — 13 · Advanced networking

1. Network services by function

Service	What it does	Signal in the wording
Azure Bastion	RDP/SSH from the portal (TLS 443) to VMs **without a public IP**; Standard SKU: native client, scaling, file transfer	"manage VMs without public IPs", "no RDP exposed"
NAT Gateway	Outbound Internet access with **fixed public IP(s)**, up to 64k SNAT ports per IP, zonal	"predictable outbound IP", "SNAT port exhaustion"
DDoS Protection	Network Protection (per VNet, all public IPs, ~3k €/month) / IP Protection (per IP); Basic free and always on	"protect public IPs from volumetric attacks", "cost-effective for few IPs" → IP Protection
Azure Firewall	Managed L3-L7 firewall; Standard (FQDN, threat intel) / Premium (TLS inspection, IDPS, URL filtering) / Basic (SMB)	"FQDN filtering", "central inspection", "TLS inspection" → Premium
Firewall Manager	Centralized firewall policies for several firewalls / secured hubs	"manage policies across hubs"
Private Endpoint	NIC with a private IP pointing to a PaaS service; blocks the public endpoint; needs a Private DNS zone privatelink.*	"no public endpoint", "access from on-prem via ER/VPN"
Private Link Service	Expose **your own service** (behind a Standard LB) to consumers in other VNets/tenants via Private Endpoint	"expose our service privately to customers"
Service Endpoint	The subnet reaches the PaaS over the backbone; the service keeps its public IP but accepts only that VNet	"restrict storage to a VNet" without requiring a private IP
DNS Private Resolver	**Hybrid** conditional DNS resolution between on-prem and Azure without DNS VMs (inbound endpoint: on-prem → Azure; outbound + ruleset: Azure → on-prem)	"on-prem must resolve private endpoints", "replace DNS VMs"
Private DNS Zones	Private zones linked to VNets; VM auto-registration	"name resolution across VNets"
Azure DNS	Public zones; alias records	"host public domain"
Route Server	**BGP** exchange between NVAs and the VNet; no manual UDRs; also with ER/VPN gateways	"NVA advertises routes dynamically"
Network Function Manager	Deploys network functions on Azure Stack Edge	Common distractor
Virtual Network Manager	Centralized management of connectivity (mesh/hub-spoke) and admin security rules at scale	"many VNets, connectivity at scale"
Network Watcher	Diagnostics: IP flow verify, NSG diagnostics, next hop, connection troubleshoot, packet capture, flow logs, topology	"allowed or denied", "why can't VM1 reach"
Traffic Analytics	Analytics over flow logs in Log Analytics	"visualize traffic patterns"
Connection Monitor	Latency/loss between endpoints (Azure, on-prem, Internet)	"monitor ExpressRoute latency"

2. ExpressRoute in detail

Concept	Detail
Peering	**Private peering** (VNets) · **Microsoft peering** (Microsoft 365, PaaS with public IP, requires NAT and route filters) · (Public peering retired)
Circuit	Redundant pair of cross-connections by design; SLA 99.95 %; SKUs Local / Standard (same geo) / **Premium** (global, >10 VNets, more routes)
ExpressRoute Direct	10/100 Gbps directly to Microsoft without a provider; supports **MACsec**
Global Reach	Connects two on-prem sites **to each other** through their ER circuits
FastPath	Traffic bypasses the gateway → lower latency and higher throughput (Ultra Performance / ErGw3AZ)
Gateway SKUs	Standard, HighPerformance, UltraPerformance, ErGw1/2/3AZ (zone-redundant)
Encryption	**Not encrypted by default**; IPsec VPN over ER (private peering) or MACsec (ER Direct)
Failover	S2S VPN over the Internet as backup ("no MPLS"); or a second circuit with another provider
ER + VPN coexistence	Same gateway subnet, VPN as backup; BGP prefers ER
Virtual WAN	ER only with Standard SKU

3. VPN Gateway in detail

Concept	Detail
Site-to-Site	IPsec/IKE; route-based (recommended, BGP, multi-site) vs policy-based (Basic, single tunnel)
Point-to-Site	SSTP (Windows only) / IKEv2 / **OpenVPN** (cross-platform, Entra ID auth); auth by **certificate** (public root uploaded to Azure, client cert on devices, revocation by thumbprint), RADIUS or Entra ID
VNet-to-VNet	Tunnel between gateways; alternative: peering (no encryption, cheaper)
Active-active	Two public IPs, two tunnels; HA
Zone-redundant SKUs	VpnGw1AZ…
SKUs	Basic (retired for new deployments, no BGP) → VpnGw1-5; throughput and tunnels grow
Gateway transit	Spoke uses the hub's gateway via peering ("Use remote gateways" / "Allow gateway transit")

Concept	Detail
Local Network Gateway	Represents the on-prem site (public IP + prefixes)
Gateway Subnet	Mandatory, /27 or larger recommended

4. Peering and routing

Rule	Detail
Peering **is not transitive**	A↔B and B↔C does not give A↔C; solutions: direct peering, hub with NVA/Azure Firewall + UDRs, Virtual WAN
Global peering	Between regions; no gateway transit towards Basic LB
UDR	Route table per subnet; next hop: VNet, Internet, Virtual appliance (IP), Virtual network gateway, None
Force traffic through the firewall	UDR with next hop = firewall private IP on the source subnets; 0.0.0.0/0 → firewall to force egress
NVA	VM with **IP forwarding** enabled; single point of failure → HA pair behind an Internal LB (HA ports) or Azure Firewall
BGP	Propagation of ER/VPN routes; "Disable BGP route propagation" on the route table so a spoke does not learn on-prem routes
Subnet sizing	Azure reserves **5 IPs**; /29 minimum (3 usable); Gateway Subnet /27; Bastion /26 (AzureBastionSubnet); Firewall /26 (AzureFirewallSubnet)
Overlap	Forbidden between connected networks (peering, VPN, ER)
Service tags	Groups of service IPs (Storage, Sql, AzureCloud) for NSG/Firewall
ASG	Group NICs for NSG rules by app role

5. Network security: what filters what

Tool	Layer	Criteria	Scope
NSG	3-4	IP, port, protocol, service tags, ASG; stateful	Subnet / NIC; most restrictive rule wins by priority
Azure Firewall	3-7	+ **FQDN**, FQDN tags, threat intel, TLS inspection (Premium)	Central hub
WAF	7	OWASP, bots, geo, rate limit	App Gateway / Front Door
DDoS	3-4	Volumetric	Public IPs
PaaS firewall	—	Allowed IPs / VNets; disable public access	Storage, SQL, Key Vault…
JIT VM access	—	Opens management ports for a limited time (Defender)	VMs

6. Private connectivity to PaaS: Private Endpoint vs Service Endpoint

	Private Endpoint	Service Endpoint
Service IP	**Private** in your VNet	Public (optimized route)
Reachable from on-prem (ER/VPN)	✓	✗
Reachable from peered VNets	✓	✗ (only the configured subnet)
Disable public endpoint	✓ possible	The service firewall restricts to the VNet
DNS	Private DNS zone required	None
Cost	Per endpoint + traffic	Free
Signal	"no public endpoint", "on-premises must access"	"restrict to VNet", minimal cost

7. Decision tree

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What do I need?
├─ Manage VMs without a public IP ................... Bastion
├─ Outbound Internet with a fixed IP ................ NAT Gateway
├─ Protect public IPs from DDoS (few IPs) ........... DDoS IP Protection (many → Network Protection)
├─ Filter by FQDN / central inspection .............. Azure Firewall (+ UDRs)
├─ Inspect TLS / IDPS ............................... Azure Firewall Premium
├─ PaaS without public endpoint, access from on-prem  Private Endpoint + Private DNS zone
├─ Expose my service privately to other tenants ..... Private Link Service
├─ On-prem resolves Azure private names ............. DNS Private Resolver
├─ NVA advertising routes dynamically ............... Route Server
├─ On-prem sites to each other over ExpressRoute .... Global Reach
├─ Encrypt ExpressRoute ............................. IPsec over ER / MACsec (Direct)
├─ ER failover over the Internet .................... S2S VPN
├─ See why a packet does not arrive ................. Network Watcher
└─ Many VNets, connectivity at scale ................ Virtual Network Manager / Virtual WAN
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8. Signals → answer

Wording	Answer
"RDP without exposing public IP"	Azure Bastion
"outbound IP must not change"	NAT Gateway

Wording	Answer
"encrypt traffic over ExpressRoute"	IPsec over ER (MACsec if ER Direct)
"connect two on-prem sites via ER"	Global Reach
"bypass the gateway for lower latency"	FastPath
"on-prem DNS must resolve Azure private endpoints"	DNS Private Resolver (inbound endpoint)
"NVA must exchange routes with BGP"	Route Server
"filter by FQDN between VNets"	Azure Firewall
"spoke must not learn on-prem routes"	Disable BGP route propagation
"packets allowed or denied"	IP flow verify / NSG diagnostics
"which route a VM uses"	Next hop
"30 VMs, avoid overlap with 172.16.0.0/16"	192.168.0.0/24
"our service consumed privately by customers"	Private Link Service

9. Mistakes to avoid

1. Treating peering as transitive.
2. Service Endpoint when access from on-prem or no public IP is required.
3. Assuming ExpressRoute is encrypted.
4. NSG to filter by FQDN.
5. A single NVA without HA.
6. Forgetting the 5 reserved IPs and the minimum sizes of special subnets.